Subject _____

Sun Mon Tue Wed Thu Fri Sat

Date:

Place

Why we Scale Attention Scores by $\sqrt{d_k}$

let, q = query vector

K = key vector

d_k = dimension of key vector

As we compute the dot product,

$$s = q \cdot K = \sum_{i=1}^{d_k} q_i k_i$$

Here, $E(q_i) = E(k_i) = 0$

so, $E(q_i k_i) = 0$

Again, $Var(q_i k_i) = E(q_i^2 k_i^2) - E(q_i k_i)^2$

$= E(q_i^2) \cdot E(k_i^2) - E(q_i) \cdot E(k_i)$

[As they are independent]

$= \{Var(q_i) + E(q_i)^2\} \cdot \{Var(k_i) + E(k_i)^2\} - 0$

$= 1$

Subject _____

Sun Mon Tue Wed Thu Fri Sat

Date:

Place

As q_i and k_i are independent of each other, their variances will add up.

$$\text{Var}(q \cdot k) = \sum_{i=1}^{d_k} \text{Var}(q_i k_i)$$

$$= d_k$$

$$\text{Standard-Deviation}(q \cdot k) = \sqrt{\text{Var}(q \cdot k)}$$

$$= \sqrt{d_k}$$

Now, softmax of a vector s is,

$$\text{softmax}(s_i) = \frac{e^{s_j}}{\sum e^{s_j}}$$

As the variance grow, one of the score dominates others, making the softmax output almost like an one-hot encoding. That is, 1 for dominating one, almost zero elsewhere. This cluster of zero gradients stop the learning process. To prevent this, we divide attention scores by $\sqrt{d_k}$ to bring the STD back to 1, preventing one hot in softmax, and to help the learning process by failitating gradient flow.